CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rhombus has perimeter 340 cm and one diagonal 26 cm. Its area (in cm²) is: 1

(a) 1092 **(b)** 2184 **(c)** 2210 **(d)** 2197

2. A ladder 17 m long, standing in a street, rests against a house on one side when its top reaches 15 m. Without moving its foot it is turned to rest against the opposite house, reaching 8 m. The width of the street (in m) is: 1

(a) 24 **(b)** 17 **(c)** 23 **(d)** 25

3. In a rectangle ABCD, P is a point inside with PA = 7, PB = 8, PC = 11. Then PD^2 equals: 1

(a) 234 **(b)** -4 **(c)** 106 **(d)** 136

4. The longest rod that fits in a room 4 m by 4 m by 7 m has length (in m): 1

(a) 15 **(b)** 9 **(c)** 8 **(d)** 10

5. The legs of a right triangle are in the ratio $7 : 24$ and its hypotenuse is 50 cm. Its perimeter (in cm) is: 1

(a) 62 **(b)** 114 **(c)** 112 **(d)** 100

6. The sides of a rectangle are in the ratio $8 : 15$ and its diagonal is 34 cm. The area of the rectangle (in cm²) is: 1

(a) 92 **(b)** 68 **(c)** 484 **(d)** 480

7. A Pythagorean triplet whose smallest member is 22 is: 1

(a) 22, 121, 123 **(b)** 22, 120, 122 **(c)** 22, 119, 121 **(d)** 22, 23, 24

8. A rhombus has perimeter 340 cm and one diagonal 72 cm. Its area (in cm²) is: 1

(a) 6120 **(b)** 2772 **(c)** 5544 **(d)** 5580

9. A ladder 25 m long, standing in a street, rests against a house on one side when its top reaches 24 m. Without moving its foot it is turned to rest against the opposite house, reaching 20 m. The width of the street (in m) is: 1

(a) 25 **(b)** 24 **(c)** 22 **(d)** 44

10. In a rectangle ABCD, P is a point inside with PA = 9, PB = 7, PC = 10. Then PD^2 equals: 1

(a) 132 **(b)** 230 **(c)** 42 **(d)** 68

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find a Pythagorean triplet in which one member is 24. 2

12. In a rectangle ABCD, P is a point inside with PA = 9 cm, PB = 7 cm, PC = 12 cm. Find PD^2. 2

13. The legs of a right triangle are in the ratio $8 : 15$ and the hypotenuse is 68 cm. Find the legs. 2

14. Find a Pythagorean triplet in which one member is 33. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 65 m long, with its foot in a street, reaches a window 63 m high on one side. Without moving its foot it is turned to reach a window 33 m high on the opposite side. Find the width of the street. 3

16. A rhombus has perimeter 340 cm and one diagonal 72 cm. Find the other diagonal and the area. 3

17. Two ships leave a port, one sailing east at 20 km/h and the other south at 21 km/h. How far apart are they after 4 hours? 3

OR

A ladder 61 m long, with its foot in a street, reaches a window 60 m high on one side. Without moving its foot it is turned to reach a window 11 m high on the opposite side. Find the width of the street. 3

18. A rhombus has perimeter 356 cm and one diagonal 78 cm. Find the other diagonal and the area. 3

OR

Two ships leave a port, one sailing east at 9 km/h and the other south at 40 km/h. How far apart are they after 2 hours? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. In quadrilateral ABCD, $\angle B = 90^\circ$, AB = 8 cm, BC = 6 cm, CD = 24 cm and DA = 26 cm. Prove that $\angle ACD = 90^\circ$ and find the area of ABCD. 5

OR

A room is 4 m long, 4 m wide and 7 m high. Find the length of the longest rod that can be placed in it. Also find the diagonal of its floor. 5

20. In quadrilateral ABCD, $\angle B = 90^\circ$, AB = 12 cm, BC = 16 cm, CD = 21 cm and DA = 29 cm. Prove that $\angle ACD = 90^\circ$ and find the area of ABCD. 5

OR

A room is 2 m long, 6 m wide and 9 m high. Find the length of the longest rod that can be placed in it. Also find the diagonal of its floor. 5

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND